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#include <stdio.h>
#include <stdlib.h>

struct node {
    int info;
    struct node *next; };

struct node *start = NULL, *temp, *follow;

void insertEnd(int value) {
    struct node *p;
    p = (struct node *)malloc(sizeof(struct node));
    p->info = value;
    p->next = NULL;
    if (start == NULL) {
        start = p;
    } else {
        temp = start;
        while (temp->next != NULL) {
            temp = temp->next; }
        temp->next = p; }
    printf("Node inserted with value %d\n", value); }

void deleteFirst() {
    if (start == NULL) {
        printf("Linked list is empty\n");
    } else {
        temp = start;
        printf("Deleting first element: %d\n", start->info);
        start = start->next;
        free(temp);}
}

void deleteLast() {
    if (start == NULL) {
        printf("List is empty.\n");}
    else if (start->next == NULL) {

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    temp = start;
    printf("Deleting last element: %d\n", start->info);
    start = NULL;
    free(temp);}
else {
    temp = start;
    while (temp->next != NULL) {
        follow = temp;
        temp = temp->next; }
    printf("Deleting last element: %d\n", temp->info);
    follow->next = NULL;
    free(temp);}
}

void deletespecificsll(int value) {
    temp = start;
    follow = NULL;
    if (start == NULL) {
        printf("List is empty.\n");
        return; }
    while (temp != NULL && temp->info != value) {
        follow = temp;
        temp = temp->next; }
    if (temp == NULL) {
        printf("%d not found\n", value);
    } else {
        if (temp == start) {
            start = start->next;
        } else {
            follow->next = temp->next; }
        printf("Deleting element: %d\n", temp->info);
        free(temp);} }

void display() {

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temp = start;
if (start == NULL) {
    printf("List is empty.\n");}
else {
    printf("Linked list elements:\n");
    while (temp != NULL) {
        printf("%d ", temp->info);
        temp = temp->next; }
    printf("\n"); }}

int main() {
    int choice, value;
    while (1) {
        printf("\n--- linked list MENU ---\n");
        printf("1. Insert at end\n");
        printf("2. Delete first element\n");
        printf("3. Delete last element\n");
        printf("4. Delete specific element\n");
        printf("5. Display list\n");
        printf("6. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("Enter value to insert: ");
                scanf("%d", &value);
                insertEnd(value);
                break;
            case 2:
                deleteFirst();
                break;
            case 3:
                deleteLast();

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        break;
    case 4:
        printf("Enter element to be deleted: ");
        scanf("%d", &value);
        deletespecificsll(value);
        break;
    case 5:
        display();
        break;
    case 6:
        exit(0);
    default:
        printf("Invalid choice!\n");
    }
}
return 0;
}
```

OUTPUT:

```
--- linked list MENU ---
1. Insert at end
2. Delete first element
3. Delete last element
4. Delete specific element
5. Display list
6. Exit
Enter your choice: 1
Enter value to insert: 10
Node inserted with value 10

--- linked list MENU ---
1. Insert at end
2. Delete first element
3. Delete last element
4. Delete specific element
5. Display list
6. Exit
Enter your choice: 1
Enter value to insert: 20
Node inserted with value 20

--- linked list MENU ---
1. Insert at end
2. Delete first element
3. Delete last element
4. Delete specific element
5. Display list
6. Exit
Enter your choice: 1
Enter value to insert: 30
Node inserted with value 30

--- linked list MENU ---
1. Insert at end
2. Delete first element
3. Delete last element
4. Delete specific element
5. Display list
6. Exit
Enter your choice: 1
Enter value to insert: 40
Node inserted with value 40

--- linked list MENU ---
1. Insert at end
2. Delete first element
3. Delete last element
4. Delete specific element
5. Display list
6. Exit
Enter your choice: 1
Enter value to insert: 50
Node inserted with value 50
```

```
--- linked list MENU ---
1. Insert at end
2. Delete first element
3. Delete last element
4. Delete specific element
5. Display list
6. Exit
Enter your choice: 1
Enter value to insert: 60
Node inserted with value 60
```

```
--- linked list MENU ---
1. Insert at end
2. Delete first element
3. Delete last element
4. Delete specific element
5. Display list
6. Exit
Enter your choice: 2
Deleting first element: 10
```

```
--- linked list MENU ---
1. Insert at end
2. Delete first element
3. Delete last element
4. Delete specific element
5. Display list
6. Exit
Enter your choice: 5
Linked list elements:
20 30 40 50 60
```

```
--- linked list MENU ---
1. Insert at end
2. Delete first element
3. Delete last element
4. Delete specific element
5. Display list
6. Exit
Enter your choice: 3
Deleting last element: 60
```

```
--- linked list MENU ---
1. Insert at end
2. Delete first element
3. Delete last element
4. Delete specific element
5. Display list
6. Exit
Enter your choice: 5
Linked list elements:
20 30 40 50
```

```
--- linked list MENU ---
1. Insert at end
2. Delete first element
3. Delete last element
4. Delete specific element
5. Display list
6. Exit
Enter your choice: 4
Enter element to be deleted: 34
34 not found

--- linked list MENU ---
1. Insert at end
2. Delete first element
3. Delete last element
4. Delete specific element
5. Display list
6. Exit
Enter your choice: 4
Enter element to be deleted: 50
Deleting element: 50

--- linked list MENU ---
1. Insert at end
2. Delete first element
3. Delete last element
4. Delete specific element
5. Display list
6. Exit
Enter your choice: 5
Linked list elements:
20 30 40

--- linked list MENU ---
1. Insert at end
2. Delete first element
3. Delete last element
4. Delete specific element
5. Display list
6. Exit
Enter your choice: 6

Process returned 0 (0x0)   execution time : 35.331 s
Press any key to continue.
```